The Director's Signature: Stylometry of Theater Choreography via Pose and Action Estimation

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Introduction

The concept of a director's "signature" or unique style is central to *auteur theory*, which posits that a director's personal vision and stylistic choices imprint a distinct identity on their work, making their contribution recognizable regardless of collaborative efforts. The notion is most commonly applied to film studies, but theater studies often holds up directors with distinct visual styles (Robert Wilson, Ivo van Hove, Romeo Castellucci) as examples of "auteurs." Unlike film, where a director has many more options in terms of camera placement, lighting, design, editing, and post-production effects, the live performance director has more limited tools and less control over the work as a whole. The main tool for a theater director is choreography of the actors' bodies and their positioning in space, a distinct phenomenon that theater scholars acknowledge yet rarely analyze. And compared to literary stylometrics, where each word and letter is generally attributable to the intentions of the author, a theater performance comprises many different mediating factors (the script, the stage design, the actors' bodies). A combined distant and narrow viewing of poses applied across multiple productions therefore is more likely to determine a director's consistent choreographic patterns, rather than only considering a single performance.

Related Work

Advances in deep-learning technologies for moving images have benefitted computational theater studies (Bardiot 2021, Escobar Varela 2021) by greatly expanding the data that can be extracted from conventional video recordings, augmenting the traditional "cinemetrics" of shot durations and pixel colors by looking deeper into the frame to identify faces, objects, poses and three-dimensional spatial relations (Chávez Heras 2023). Applying such tools across many recordings of staged performances further enables new "distant viewing" analyses (Arnold / Tilton 2019), including efforts to quantify and compare stage directors' "signature" choreographic styles. The body of scholarship on literary stylometry offers promising questions and methods, although our current effort ventures into non-linguistic realms rarely visited even in previous "beyond authorship" studies of theatrical style that focused primarily on the written word (Craig / Greatley-Hirsch 2017). Our work owes much to efforts to categorize deployments of human poses in visual art history (Kutrzyński / Król 2024), although considering theatrical performances greatly increases the degree of spatio-temporal complexity.

Corpus and Data Preprocessing

We assembled a corpus of 30 full-length theatrical performance recordings, evenly divided among three contemporary "auteur" directors: Bill T. Jones, Romeo Castellucci, and Krzysztof Warlikowski (Appendix 1). We applied an advanced transformer-based pose estimation and tracking tool (Rajasegaran et al. 2022) to each recording, generating both 2D and estimated 3D pose coordinates of figures tracked across consecutive frames. We reduced each estimated pose to the 13-keypoint COCO standard (head, shoulders, hips, arms, legs) for compatibility with downstream tools and also divided each pose track into "movelets" of 1/6 of a second for motion quantification. We further applied an action recognition model that assigns a 60-element Atomic Visual Actions (AVA) embedding vector to each individual pose (Rajasegaran et al. 2023), and also generated a 16-element probabilistic view-invariant "POEM" embedding for each pose (Sun et al. 2020).

The data described above were loaded into an instance of the MIME (Machine Intelligence for Motion Exegesis) framework (github.com/sulcidr/mime), which provides a web-based video pose timeline exploration interface and is built upon a Postgres database using the pgvector extension to enable fast similarity searching of pose and action vectors and coordinates (Figure 1).

Figure 1. The MIME interface's timeline view, displaying pose and motion data for a full performance, with details and inferred 3D body armatures for a selected frame.

We queried the MIME instance's API to derive a set of pose, action, motion and distance statistics for each performance. Averages (mean and median) and standard deviation of each distribution were calculated per populated frame for the elements of the following:

- POEM pose embedding values
- AVA action recognition embedding values
- 3D "global orientation" representations of each pose (13 x 3 coordinates in meters)
- 3D distances between pose centroids (meters)
- In-place keypoint movement of each tracked pose movelet (meters/second)
- "Sidereal" motion (relative to the background) of each tracked pose movelet (meters/second)

Examining multiple performances per director helps to attenuate the influence of "noise" data from non choreography-related shots of musicians, audience members and post-performance curtain calls (for example), as well as the vision model's struggles with certain camera effects and lighting conditions, and perhaps most significantly, choices of shot angles and framing made by the video directors, who are usually not the same person as the stage director.

Analytical Methods

We framed our primary stylometric inquiry as a performance-to-director classification problem: from the derived pose, action, motion and distance data above, which elements are most effective at predicting the correct stage director for a given performance? We began with "leave one out" (LOO) tests in which we assembled the average pose distance and motion features into a vector for each performance, repeated the process for the 3D "global orientation" pose keypoint coordinates, view-invariant POEM pose vectors and action recognition vectors, and compared these to aggregate vectors that averaged values from all of a director's works except the held-out performance. Simple cosine similarity-based matching of directors to held-out works gave way to machine-learning classification algorithms (a "random forest" combined decision-tree estimator was slightly more accurate than others), which facilitated tests to estimate the relative contribution ("importance") of each element of the feature vectors to successful predictions. Tests were repeated with multiple random seeds and eventually via 10-fold cross-validation to avoid inadvertently cherry-picking results.

	Motion & distance	**Pose embeddings**	**3D global pose keypoints**	**Action embeddings**
LOO cosine similarity	63.33% (19/30)	80% (24/30)	73.33% (22/30)	80% (24/30)
LOO random forest	61% (183/300)	89% (267/300)	66% (198/300)	80.67% (242/300)
10-fold random forest	58.7% stdev: 35%	68% stdev: 35.3%	68.80% stdev: 32.3%	71.60% stdev: 32.1%

Table 1. Accuracies of performance-to-director classifiers by method (rows) and input features (columns). 33% accuracy is expected by chance. Input features are based on mean values of component elements across performances (median values resulted in slightly lower accuracies).

Figure 2. Confusion matrices of directorial classifications of 30 performances using a "leave one out" train/test split across 10 runs with different random seeds.

Figure 3. Feature importances from "random forest" classifiers based on motion and distance (left) and pose embeddings (right).

Results

As Table 1 and Figure 2 (left) indicate, distance and motion statistics aggregated across each performance helped to discern stagings by Bill T. Jones from the others, but the confusion between Castellucci and Warlikowski suggests this may be seizing upon the differences in *genre* between their oeuvres, rather than specific directorial styles: the performances in our corpus directed by Jones are dance works, while the others' tend to be from more static genres, particularly opera. Yet comparing average pose embeddings was largely successful at differentiating between all three directors (see Figure 2, right). Average action embeddings performed similarly (Table 1). This may be due to the tendency of embedding spaces to accentuate semantic differences that are "baked in" during pre-training – which also suggests why comparing the raw global 3D pose data was not as successful, as it does not incorporate such learned biases. The drop-off in accuracy of the embedding-based classifiers during 10-fold cross-validation does however imply that they struggle when the training set is reduced.

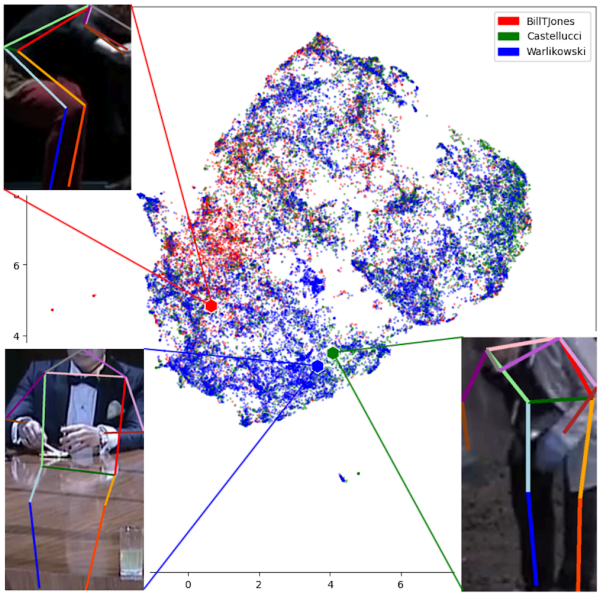

Figure 4. UMAP projection of the POEM embeddings of a 1% sample of poses in the corpus, color-coded by director, showing the locations of the average embedding for each director and sample poses with high similarity to these embeddings.

The superior performance overall of the embedding-based approaches, however, has the drawback that it is more difficult to translate their computational discernment into actual insights regarding the directors' favored poses and action patterns. It is not generally possible to convert an averaged POEM embedding to a single pose, nor is it straightforward to determine precisely what the importance of "feature 1" (Figure 3, right) means for actual poses. Likewise, the AVA action embeddings only map to fairly vague descriptions such as "51%: stand, 38%: watch a person, 33%: carry an object." Yet thanks to the massive vector database available via the MIME platform, it is at least possible to explore poses from the full data set whose embeddings are most similar to a director's averages. Figure 4 suggests this can reveal useful insights, such as the overall greater similarity between Castellucci's and Warlikowski's pose repertoires relative to Jones's, and the tendencies of Jones to employ dramatically contorted poses, Castellucci to favor hunched-over standing postures, and Warlikowski to deploy figures in an active sitting position.

Conclusion

This work demonstrates technologies and methods for addressing the question of directorial style in theater via distant and close viewing of pose choreography. Future iterations will introduce a more diverse set of performances and directors and investigate whether action embeddings can help to untangle the computational challenges inherent to quantification of gesture. Other enhancements will better control for genre and content by comparing different directors' stagings of the same work, and consider scene-level subdivisions of the source performances.

Table 2. Details of the 30 full-length performance recordings analyzed.

Title	Director	Duration	Poses	Tracks	Shots
A Letter to My Nephew	Bill T. Jones	1:09:02	308088	698	182
Analogy/Dora: Tramontane	Bill T. Jones	1:17:12	411046	2136	1794
A Quarrelling Pair	Bill T. Jones	1:17:50	267257	759	729
D-Man in the Waters	Bill T. Jones	0:35:10	141191	855	110
Fondly Do We Hope... Fervently Do We Pray	Bill T. Jones	1:23:55	435160	1001	846
Holzer Duet... Truisms	Bill T. Jones	0:17:45	43206	88	65
Play and Play	Bill T. Jones	1:12:50	462918	2359	230
Secret Pastures	Bill T. Jones	1:33:54	419080	1470	14
Story/Time: The Life of an Idea	Bill T. Jones	1:13:35	402026	1470	869
We Shall Not Be Moved	Bill T. Jones	1:57:30	633482	3244	4111
Das Floß der Medusa (Henze)	Castellucci	1:21:35	114133	724	549
Democracy in America	Castellucci	1:43:05	219148	524	46
Don Giovanni (Mozart)	Castellucci	3:30:01	897915	5155	1009
Go Down, Moses	Castellucci	1:02:40	122960	289	78
Inferno	Castellucci	1:36:16	345711	2837	799
Parsifal (Wagner)	Castellucci	3:58:29	780960	3551	847
Purgatorio	Castellucci	1:13:48	93387	399	392
Requiem (Mozart)	Castellucci	1:38:57	507498	2874	365
Resurrección (Gustav Mahler)	Castellucci	1:40:15	295840	2212	610
The Magic Flute (Mozart)	Castellucci	2:43:15	763697	4385	863
Bluebeard's Castle (Bartók) / La voix humaine (Poulenc)	Warlikowski	2:02:19	278984	1389	736
Die Gezeichneten (Schreker)	Warlikowski	3:48:41	954114	5454	1345
Elektra (Strauss)	Warlikowski	2:06:25	324719	1220	532
Iphigénie en Tauride (Gluck)	Warlikowski	2:11:26	510703	2648	822
Lady Macbeth of Mtsensk (Shostakovich), 1	Warlikowski	1:47:13	412434	3402	774
Lady Macbeth of Mtsensk (Shostakovich), 2	Warlikowski	1:17:36	446255	3619	526
Médée (Cherubini)	Warlikowski	2:24:41	653184	4443	1029
Les Contes d'Hoffmann	Warlikowski	3:08:39	930245	6181	1432

(Offenbach)					
The French	Warlikowski	3:45:47	906390	3410	982
Wozzeck (Berg)	Warlikowski	1:45:20	352588	2766	887

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